Table 1: Type mapping. Each C++ type maps to the Lean type of the same representation, so that arithmetic on both sides is the same IEEE 754 or two's-complement operation.

C++, after typedefs	Lean
`float`, `double`, `bool`	`Float32`, `Float`, `Bool`
`int8_t` … `int64_t`; `int`	`Int8` … `Int64`; `Int32`
`uint8_t` … `uint64_t`; `size_t`	`UInt8` … `UInt64`; `USize`
`T[N]`, N a constant	`Array T` of N elements, checked access
plain `struct S`	`structure S`
`class C`	`C.State`, `C.Params`, `C.Inputs`, `C.Outputs`
`enum E`	`inductive E` with `E.toInt` and `E.ofInt`
parameter wrapper `P<T>`	a field of `C.Params` of type `T`
bit-field `bool x : 1`	`Bool`